Purpose of the Lab

The purpose of this lab is to introduce core concepts in robotic sensing and navigation using GPS data and ROS. Students collect and parse GPS data, convert latitude and longitude to UTM coordinates, and publish the results using a custom ROS message. The lab also involves recording data with ROS bag files by echoing topic "/gps" and applying statistical analysis to evaluate GPS performance during stationary and walking conditions, while identifying sources of measurement error.

Stationary Data Sets

Scatter plots of Northing vs Easting data for cluttered and open data sets

And Altitude vs Time scatter plots for cluttered and open datasets

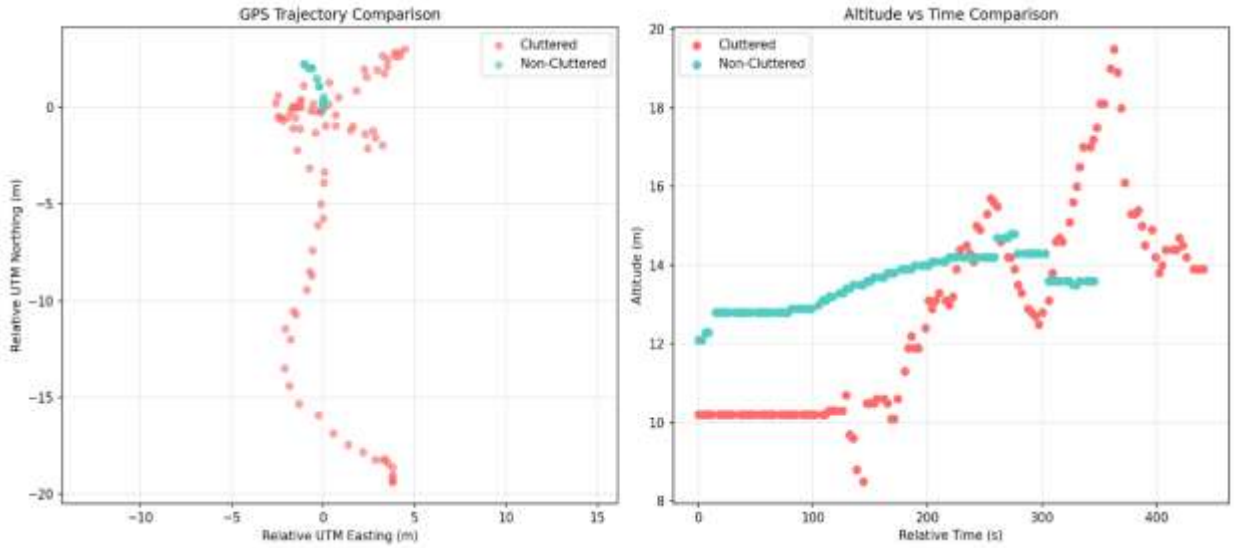


Fig1: Plots for cluttered and open data sets

Error estimates for Cluttered and Open data sets

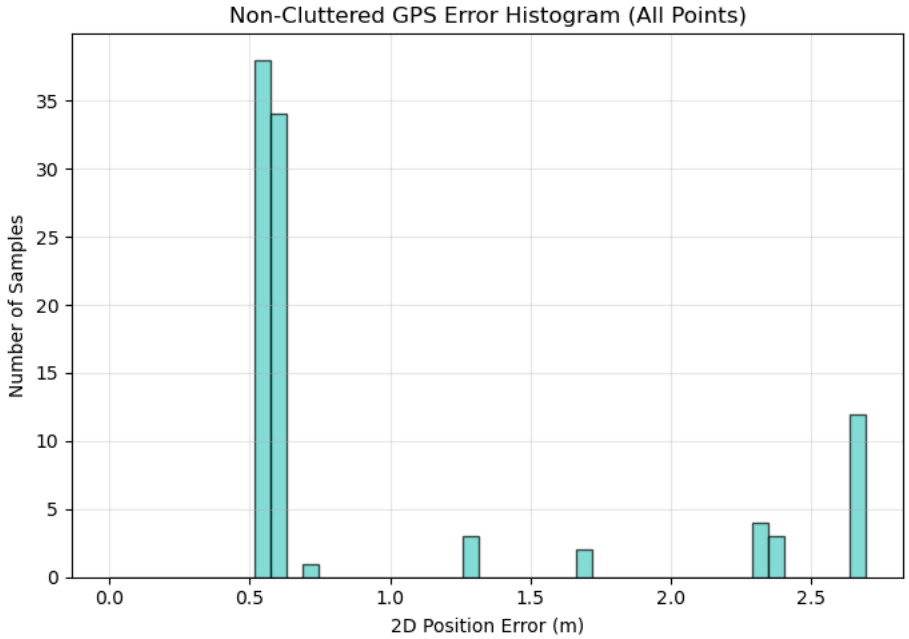


Fig2: Histogram of error from known position for open data

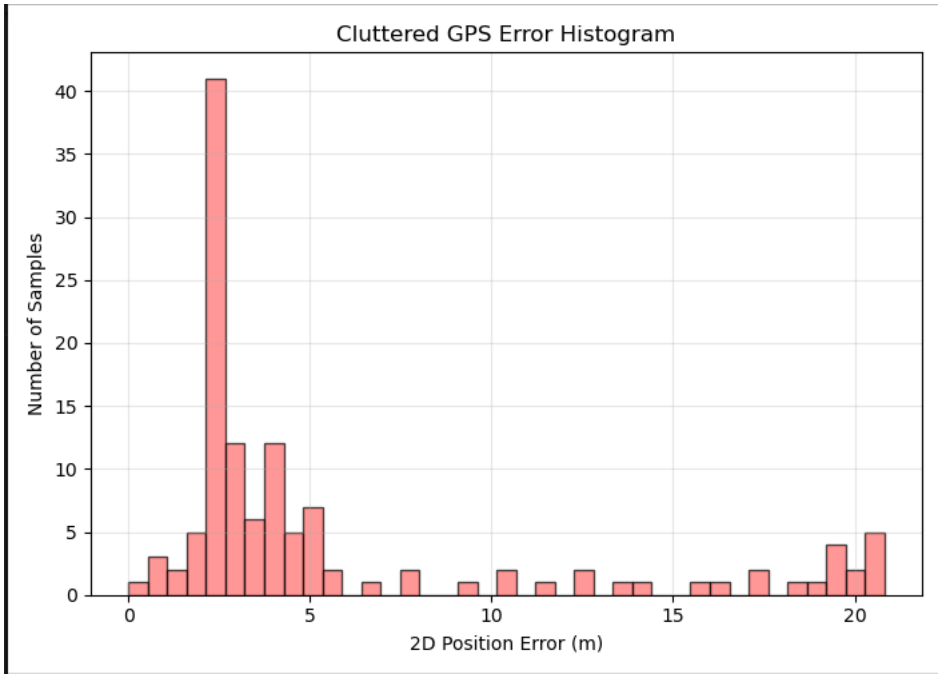


Fig3: Histogram of error from known position for cluttered data

Error estimate quantification:

```
Cluttered Errors (known reference):
Reference position: Easting=328330.000, Northing=4689545.000
Mean error:   5.933 m
Median error: 3.074 m
95% error:    19.837 m

Non-Cluttered Errors (known reference):
Reference position: Easting=327984.149, Northing=4689511.170
Mean error:   1.005 m
Median error: 0.602 m
95% error:    2.693 m
```

Fig4: Error estimate(mean-error) for cluttered and open data

In the non-cluttered (open-sky) environment, the mean positional error of approximately 1 m is reasonable. Low HDOP values(almost 1) indicate good satellite geometry, which typically results in higher positional accuracy.

In contrast, the cluttered environment exhibits a much larger mean error of approximately 5 m, which is also expected. Higher HDOP values(almost 3) in cluttered environments reflect poor satellite geometry caused by obstructions such as buildings or trees. Also, due to multipath effect it causes the error to be on the higher side. One more factor is in the cluttered area there was an offset in the reference co-ordinates on Google earth causing error to be high.

Straight-line dataset Analysis

Scatter plot of Northing vs Easting plot for moving data

Altitude vs time plot for moving data

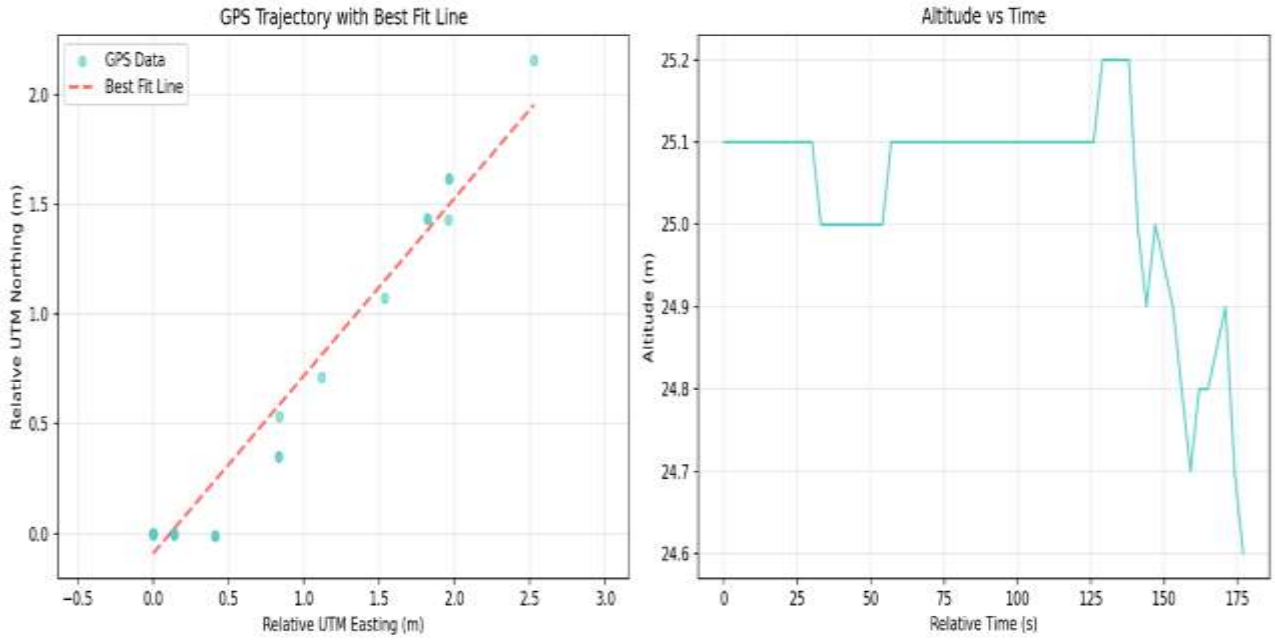


Fig5: Plots for moving data

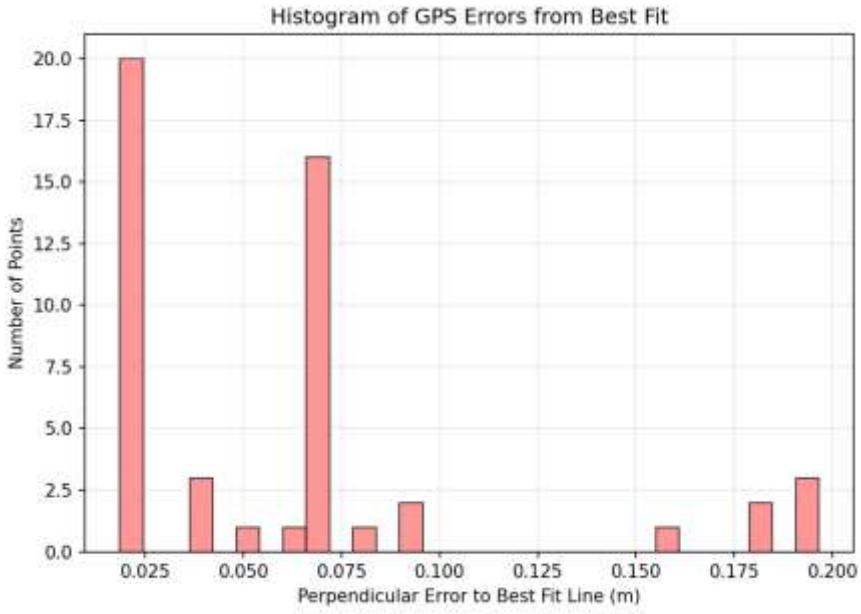


Fig6: Error estimate for moving data

Error quantification for moving data:

```
Processing 50 GPS messages...
Extracted 50 GPS points (failed: 0)
  Time range: 1770312827.98 to 1770313004.98
  UTM Easting range: 328114.75 to 328117.27
  UTM Northing range: 4689334.18 to 4689336.35
  Altitude range: 24.60 to 25.20
Data saved to 'gps_data.csv'

Best fit line: y = 0.81020 * x + -0.09085
Mean perpendicular error from best-fit line: 0.062 m
```

Fig7: Error estimate quantification (mean error)

Mean_error for moving data from best-fit line = 0.062m

Questions:

a. How do you estimate the error values for stationary vs. moving data?

For the stationary data, the error is estimated by computing the distance between each recorded GPS position and a fixed reference coordinate, and then taking the mean of these distances. This approach calculates the positional error of the GPS puck

For the moving data, the error is estimated relative to a best-fit line computed from the recorded trajectory. The perpendicular distance of each GPS point from this best-fit line is calculated, and the mean of these distances is used to quantify mean error.

b. Why does this result occur, and what does this say about GPS navigation while moving?

The stationary error reflects GPS puck accuracy and is strongly affected by environmental conditions and reference position accuracy (which is in this case was lower for cluttered space), leading to larger errors in cluttered environments. In contrast, the moving-data error is smaller because it measures deviation from a fitted path, which removes constant bias in the GPS measurements.

When comparing the mean error for moving data is less as compared to the stationary data as the mean error of stationary data set is calculated based n a known reference (the value of which might be skewed) and also the mean error of moving data is calculated based off the best-fit line. This indicates that even if the GPS puck has high mean error from the known position it can still give a correct relative positioning.

c. What are physically likely source(s) of error in these data sets?

Likely sources of error include:

- Multipath effects and signal blockage in cluttered environments
- Poor satellite geometry (high HDOP)
- Offset in reference coordinates

- Receiver noise and measurement jitter
- Atmospheric factors- Because different data sets were collected at different times of the day.